COLLEGE OF
ENGINEERING AND
COMPUTER SCIENCE

CCDS-322 PROJECT

PROJECT

TRAFFIC CONGESTION
PREDICTION
USING MACHINE LEARNING
AN IOT-BASED APPROACH

Presented To

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INTRODUCTION

Traffic congestion is an increasing problem that impacts urban efficiency, the environment, and daily life. With rapid urbanization and the growing number of vehicles on the roads, predicting congestion has become essential for improving traffic management and reducing delays. Machine learning techniques provide a powerful way to model and predict congestion using real-world data. By analyzing spatial and temporal features such as location, traffic density, and time of day, predictive models can help authorities take proactive actions to manage traffic flow.

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INVESTIGATION AND DATA SOURCES

2.1 Project Goal

The main goal of this project is to develop machine learning models capable of predicting traffic congestion levels using traffic sensor data.

The project aims to:

- Analyze traffic-related features.
- Predict congestion levels (Low, Medium, High).
- Compare different machine learning algorithms.
- Identify the best-performing model based on accuracy and validation results.

2.2 Dataset Description and Rationale

The dataset used in this study, titled 'IoT_Traffic Dataset', was obtained from Kaggle

Column Name	Description
Latitude, Longitude	Geographic coordinates of the location
Traffic_Speed_kmph	Traffic speed in kilometers per hour
Traffic_Density_vpkm	Traffic density
FFT_Feature_1,FFT_Feature_2, FFT_Feature_3	Features extracted using Fast Fourier Transform (FFT).
Vehicle_Count	Number of vehicles in the area
Time_of_Day_min	Time in minutes of the day
Congestion_Level	Level of congestion (0: Low, 1: Medium, 2: High)

Reasons for choosing this dataset

1. **It contains multiple variables that influence traffic conditions, enabling analysis of complex relationships.**
2. **It provides data suitable for supervised classification tasks.**
3. **It is clean, structured, and sufficiently large for model training and validation.**

2.3 Hypotheses and Research Questions

Hypothesis 1: Higher traffic density leads to higher congestion levels.

Hypothesis 2: Lower traffic speeds indicate higher congestion.
Research Questions:

- How do traffic speed and density affect the congestion level?
- Can predictive models provide accurate predictions for different congestion levels?
- Which machine learning algorithm performs best for this prediction task?

2.4 Tools and Libraries Used

The project was implemented using Python in the Google Colab environment. The following libraries were used:

Library	Purpose
Pandas	For data analysis and cleaning.
NumPy	For performing numerical operations
Matplotlib & Seaborn	For creating visualizations
Scikit-learn (sklearn)	For data analysis and machine learning algorithms.

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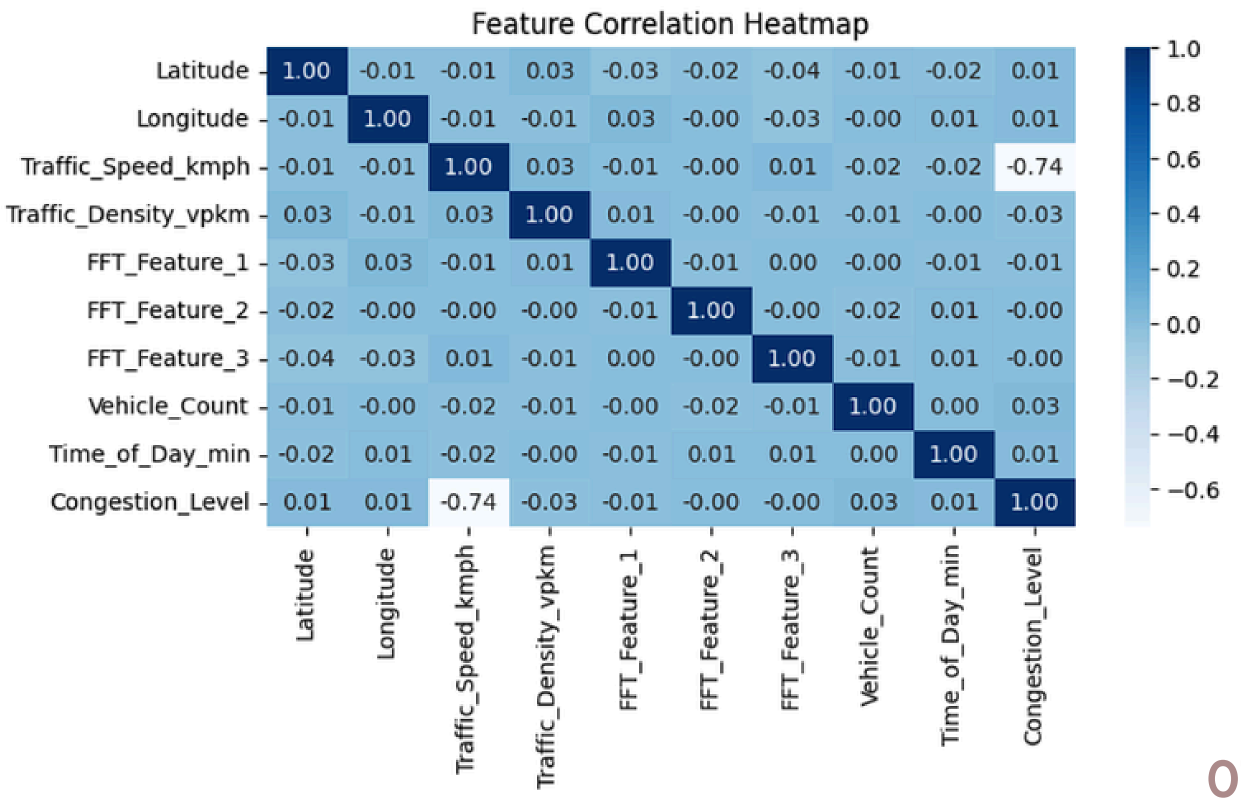
MODELS AND RESULTS

3.1 Data Preparation and Preprocessing

Before applying the machine learning models, the dataset was prepared and analyzed to ensure reliable model performance. The dataset contained 3,876 records and 10 features, including traffic speed, traffic density, FFT features, vehicle count, and time-related information. No missing values were detected in the dataset, eliminating the need for data imputation or additional cleaning procedures.

A correlation analysis was then performed to examine relationships between variables and identify highly correlated features.

Feature Correlation Heatmap



3.1 Data Preparation and Preprocessing

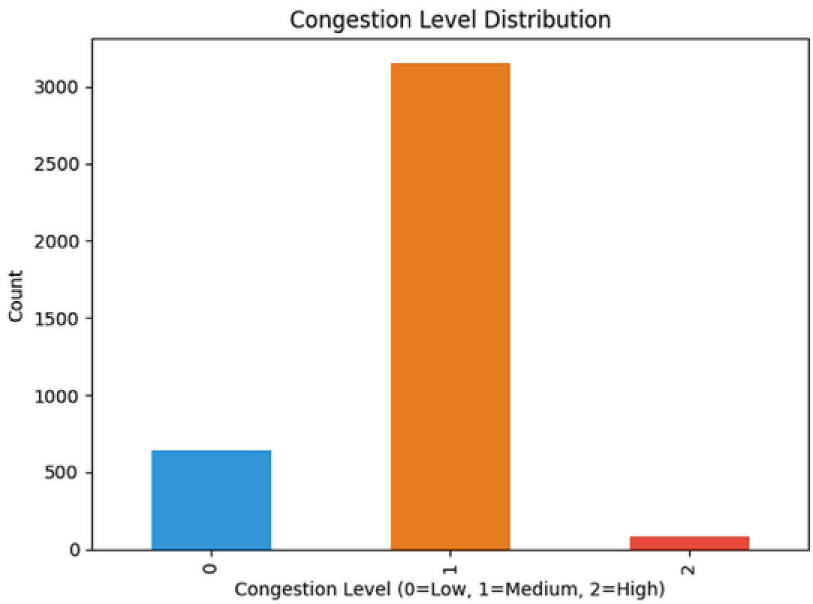
The heatmap revealed a strong negative correlation between Traffic_Speed_kmph and Congestion_Level (approximately -0.74), indicating that lower traffic speed is associated with higher congestion levels. To avoid data leakage and improve model reliability, highly correlated traffic-related features were excluded from the training process.

The final selected features used for modeling were:
Latitude , Longitude , FFT_Feature_1 , FFT_Feature_2 , FFT_Feature_3 , Time_of_Day_min
The target variable, Congestion_Level, consisted of three classes:

Low Congestion / Medium Congestion / High Congestion

However, the dataset showed significant class imbalance, with the "Medium" congestion category representing the majority of samples.

Congestion Level Class Distribution




```
Congestion_Level
1    3151
0     644
2      81
Name: count, dtype: int64
```

To maintain fair class representation during model training and testing, the dataset was divided using stratified sampling into:

- 80% training data
- 20% testing data

Finally, feature scaling was applied using StandardScaler to normalize feature ranges and improve model performance, particularly for Logistic Regression.

3.2 Applied Machine Learning Algorithms

Three supervised machine learning algorithms were implemented and compared to evaluate traffic congestion prediction performance.

1 Logistic Regression

Logistic Regression was used as a baseline linear classification model.

Model Settings:

LogisticRegression(max_iter=1000, class_weight='balanced')

Results:

- Test Accuracy: 29.9%
- Cross Validation Accuracy: 30.5%

The model struggled to capture non-linear relationships and showed poor classification performance.

2 Decision Tree Classifier

Decision Tree is a non-linear classification model capable of learning hierarchical decision rules.

Model Settings:

DecisionTreeClassifier(random_state=42, class_weight='balanced')

Results:

- Test Accuracy: 67.27%
- Cross Validation Accuracy: 68.55%

The model performed significantly better than Logistic Regression and captured non-linear patterns more effectively.

3 Random Forest Classifier

Random Forest is an ensemble learning algorithm based on multiple decision trees.

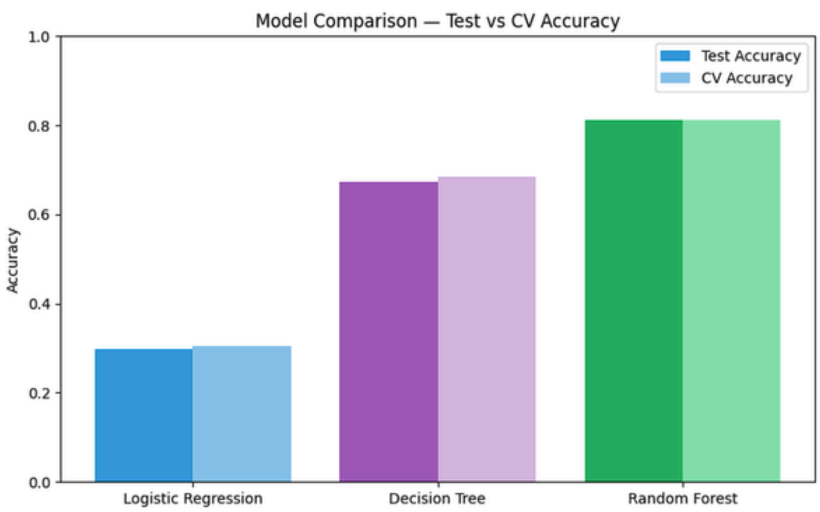
Model Settings:

RandomForestClassifier(n_estimators=100, random_state=42)

Results:

- Test Accuracy: 81.19%
- Cross Validation Accuracy: 81.16%
- Random Forest achieved the best overall performance among all models.

3.3 Model Results and Comparison (Test vs CV Accuracy)



Model	Test Accuracy	CV Accuracy
Logistic Regression	0,299	0,305
Decision Tree	0,673	0,686
Random Forest	0,812	0,812

The results indicate that ensemble methods outperform linear models for this traffic congestion classification problem. However, all models struggled to correctly classify the minority “High” congestion class due to severe class imbalance.

3.4 Model Validation

Several validation techniques were applied to ensure the reliability, consistency, and generalization capability of the developed machine learning models. Initially, the train-test split method was used to evaluate model performance on previously unseen testing data. In addition, 5-Fold Cross Validation was performed to assess model stability across multiple data partitions and reduce the risk of biased evaluation results.

Confusion matrices were also generated for each model to visualize classification outcomes and identify prediction errors between congestion categories. Furthermore, multiple evaluation metrics, including Accuracy, Precision, Recall, and F1-Score, were used to provide a more comprehensive assessment of model performance, particularly considering the imbalanced nature of the dataset.

Across all validation methods, the Random Forest model consistently demonstrated the strongest predictive performance and maintained stable accuracy with minimal variance between folds. These results indicate that Random Forest provided the best balance between predictive accuracy, robustness, and generalization capability for the traffic congestion prediction task.

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SUMMARY AND FINDINGS

4.1 Summary and Results

This project successfully demonstrated the effectiveness of machine learning in predicting traffic congestion levels — Low, Medium, and High — using spatial, temporal, and engineered features. Three supervised models were implemented and evaluated with the following results:

Model Performance:

- Random Forest → 81.19% accuracy (Best Model)
- Decision Tree → 67.27% accuracy
- Logistic Regression → 29.90% accuracy

Key Findings:

- Ensemble methods significantly outperform linear models for traffic congestion classification.
- All models struggled to correctly identify the High Congestion class due to severe class imbalance, where Medium class dominated with 3,151 samples compared to only 81 High congestion samples.
- Traffic speed showed a strong negative correlation (−0.74) with congestion level, confirming that lower speeds are associated with higher congestion.
- The dataset was split using stratified sampling (80% training, 20% testing) to maintain fair class representation.
- Feature scaling was applied using StandardScaler to normalize feature ranges and improve model performance.

Model Performance Comparison

Model	Test Accuracy	Cross Validation
Logistic Regression	29,90%	30,52%
Decision Tree	67,27%	68,55%
Random Forest	81,19%	81,16%

Congestion Level Distribution

As shown earlier in Section 3.1, the dataset was highly imbalanced with Medium congestion dominating (3151 samples), compared to Low (644) and High (81) which directly impacted model performance on minority classes.

4.1 Summary and Results (Continued)

Despite the overall strong performance, all models showed difficulty in correctly identifying the High Congestion category. This limitation was primarily caused by the class imbalance problem, where the number of samples in the "High congestion" class was significantly smaller than those in other categories. As a result, the models tended to favor the dominant Medium Congestion class during prediction.

Distribution of Traffic Congestion Classes

--- Logistic Regression ---

	precision	recall	f1-score	support
Low	0.15	0.24	0.19	129
Medium	0.80	0.31	0.45	631
High	0.01	0.25	0.02	16
accuracy			0.30	776
macro avg	0.32	0.27	0.22	776
weighted avg	0.68	0.30	0.40	776

--- Decision Tree ---

	precision	recall	f1-score	support
Low	0.18	0.20	0.19	129
Medium	0.81	0.78	0.80	631
High	0.05	0.06	0.05	16
accuracy			0.67	776
macro avg	0.35	0.35	0.35	776
weighted avg	0.69	0.67	0.68	776

--- Random Forest ---

	precision	recall	f1-score	support
Low	0.33	0.01	0.02	129
Medium	0.81	1.00	0.90	631
High	0.00	0.00	0.00	16
accuracy			0.81	776
macro avg	0.38	0.33	0.30	776
weighted avg	0.72	0.81	0.73	776

Best Model: Random Forest

Test Accuracy: 81.19%

CV Accuracy: 81.16%

To further analyze model behavior, the confusion matrix of the best-performing model (Random Forest) was examined. The matrix shows strong prediction accuracy for the majority classes, but weaker performance in detecting rare high-congestion events.

Confusion Matrices for All Models

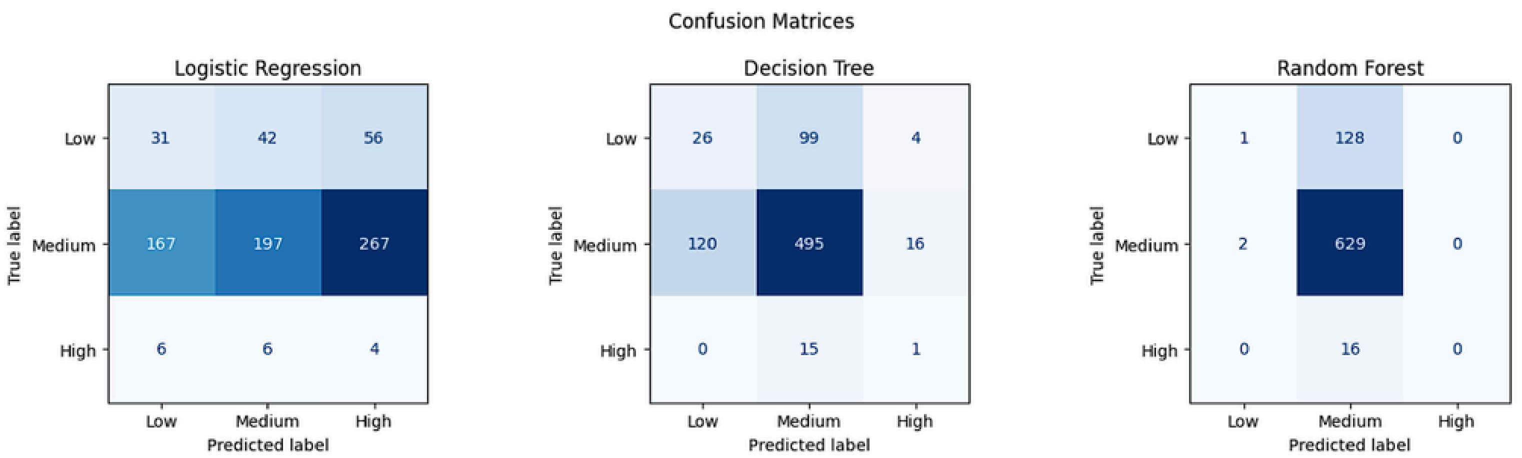

Figure 4:The matrices show the predicted versus actual congestion levels for each model.

Key Takeaway:

Machine learning models, particularly ensemble methods such as Random Forest, proved highly effective for predicting general traffic conditions. However, additional improvements are still needed to better capture rare but critical high congestion scenarios, which are essential for building reliable real-world intelligent traffic management systems

4.2 Recommendations (Future Work)

To improve prediction accuracy and enhance the practical value of the system, the following recommendations are proposed for future development:

1. Address Class Imbalance
2. Apply oversampling techniques such as SMOTE, or collect more data for minority classes to improve prediction of rare congestion events.
3. Add More Features
4. Incorporate additional traffic-related factors such as:
    - Weather conditions
    - Traffic accidents
    - Day of the week
    - Public events and holidays
5. Optimize Model Parameters
6. Use hyperparameter tuning techniques such as Grid Search to maximize model performance.
7. Test Advanced Models

8. **Explore more advanced machine learning models such as:**
   - **Gradient Boosting**
   - **XGBoost**
   - **Neural Networks**
9. **Real-World Deployment**
10. **Integrate the trained model into an intelligent transportation dashboard and monitor its real-time performance.**

4.3 Ethical and Professional Responsibilities

Developing AI-based transportation systems requires attention to ethical and professional responsibilities to ensure fairness, safety, and reliability.

1. **Data Privacy**

   Traffic data must be anonymized and handled securely to protect user and location information.

2. **Bias and Fairness**

   Unbalanced datasets may cause biased predictions, so models should be regularly tested for fairness.

3. **Transparency**

   Prediction methods should be understandable. Simple models are easier to interpret, while complex models may need explainability tools.

4. **Reliability and Safety**

   Accurate predictions are important to avoid unsafe traffic decisions, requiring continuous monitoring and evaluation.

5. **Accountability**

   Developers and authorities are responsible for system outcomes, and human supervision should remain part of decision-making.

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CONCLUSIONS

This project successfully demonstrated the potential of machine learning in solving real-world traffic congestion prediction problems. By evaluating three models — Logistic Regression, Decision Tree, and Random Forest — it was proven that ensemble methods, particularly Random Forest, deliver the most reliable and accurate predictions. While the results were promising, challenges such as class imbalance and limited features highlighted the need for further improvements. With future enhancements including better data collection, advanced modeling techniques, and real-world deployment, this system has the potential to significantly contribute to the development of intelligent and efficient traffic management systems.

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